

1. Express each of the following as a single trigonometric ratio, and then evaluate the exact value of the ratio.

a)  $\sin^2\left(\frac{7\pi}{3}\right) - \cos^2\left(\frac{7\pi}{3}\right)$

$$= -\cos\left(\frac{14\pi}{3}\right)$$

$$= -\cos\left(5\pi - \frac{\pi}{3}\right)$$

$$= \cos\left(\frac{\pi}{3}\right)$$

$$= \frac{1}{2}$$

b)  $1 - 2\sin^2\left(\frac{5\pi}{12}\right)$

$$= \cos\left(\frac{5\pi}{6}\right)$$

$$= -\cos\left(\frac{\pi}{6}\right)$$

$$= -\frac{\sqrt{3}}{2}$$

c)  $\sin\left(\frac{5\pi}{6}\right)\cos\left(\frac{3\pi}{8}\right) - \cos\left(\frac{5\pi}{6}\right)\sin\left(\frac{3\pi}{8}\right)$

$$= \sin\left(\frac{5\pi}{6} - \frac{3\pi}{8}\right)$$

$$= \sin\left(\frac{11\pi}{24}\right)$$

$$\cos\left(\frac{11\pi}{12}\right) = 1 - 2\sin^2\left(\frac{11\pi}{24}\right)$$

$$\cos\left(\frac{11\pi}{12}\right) = \cos\left(\pi - \frac{\pi}{12}\right)$$

$$= -\cos\left(\frac{\pi}{12}\right)$$

$$= -\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right)$$

$$-\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right) = 1 - 2\sin^2\left(\frac{11\pi}{24}\right)$$

$$2\sin^2\left(\frac{11\pi}{24}\right) = 1 + \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$= \frac{4 + \sqrt{6} + \sqrt{2}}{4}$$

$$\sin^2\left(\frac{11\pi}{24}\right) = \frac{4 + \sqrt{6} + \sqrt{2}}{8}$$

$$\sin\left(\frac{11\pi}{24}\right) = \frac{\sqrt{4 + \sqrt{6} + \sqrt{2}}}{2\sqrt{2}}$$

$$\sin\left(\frac{11\pi}{24}\right) = \frac{\sqrt{4\sqrt{2} + 2\sqrt{3} + 2}}{4}$$

$$\text{d) } 2 \cos^2\left(\frac{5\pi}{8}\right) - 1$$

$$= \cos\left(\frac{5\pi}{4}\right)$$

$$= \cos\left(\pi + \frac{\pi}{4}\right)$$

$$= -\cos\left(\frac{\pi}{4}\right)$$

$$= -\frac{\sqrt{2}}{2}$$

$$\text{e) } \cos\left(\frac{4\pi}{3}\right) - \csc\left(\frac{5\pi}{6}\right)$$

$$= \cos\left(\pi + \frac{\pi}{3}\right) - \csc\left(\pi - \frac{\pi}{6}\right)$$

$$= -\cos\left(\frac{\pi}{3}\right) - \frac{1}{\sin\left(\frac{\pi}{6}\right)}$$

$$= -\frac{1}{2} - 2$$

$$= -\frac{5}{2}$$

2. If  $\cos(\alpha) = \frac{4}{5}$ , and  $\sin(\beta) = -\frac{12}{13}$ ,  $0 < \alpha < \frac{\pi}{2}$ ,  $\pi < \beta < \frac{3\pi}{2}$ , evaluate  $\cos(2\alpha + \beta)$ .

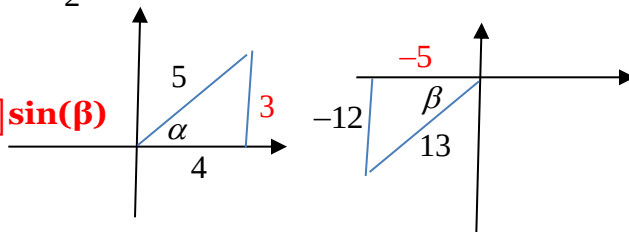
$$\cos(2\alpha + \beta) = \cos(2\alpha)\cos(\beta) - \sin(2\alpha)\sin(\beta)$$

$$= [2\cos^2(\alpha) - 1]\cos(\beta) - [2\sin(\alpha)\cos(\alpha)]\sin(\beta)$$

$$= \left[2\left(\frac{4}{5}\right)^2 - 1\right]\left(\frac{-5}{13}\right) - \left[2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right)\right]\left(\frac{-12}{13}\right)$$

$$= \left(\frac{7}{25}\right)\left(\frac{-5}{13}\right) + \frac{288}{325}$$

$$= \frac{253}{325}$$



3. If  $\frac{\pi}{2} < x < \pi$  and  $\cos^2(x) = \frac{8}{9}$ , determine the **exact** value of  $\sin(4x)$ .

$$\sin(x) = \sqrt{1 - \cos^2(x)} \quad , \quad \cos(x) = -\frac{2\sqrt{2}}{3}$$

$$= \sqrt{1 - \frac{8}{9}}$$

$$= \frac{1}{3}$$

$$\sin(4x) = 2\sin(2x)\cos(2x)$$

$$= 4\sin(x)\cos(x)[2\cos^2(x) - 1]$$

$$= 4\left(\frac{1}{3}\right)\left(-\frac{2\sqrt{2}}{3}\right)\left[2\left(\frac{8}{9}\right) - 1\right]$$

$$= \left(-\frac{8\sqrt{2}}{9}\right)\left(\frac{7}{9}\right)$$

$$= -\frac{56\sqrt{2}}{81}$$